Intelli-Credit

AI-Powered Corporate Credit Decisioning Engine

Resolving the 'Data Paradox' for Indian Corporate Lending

The Problem: The 'Data Paradox'

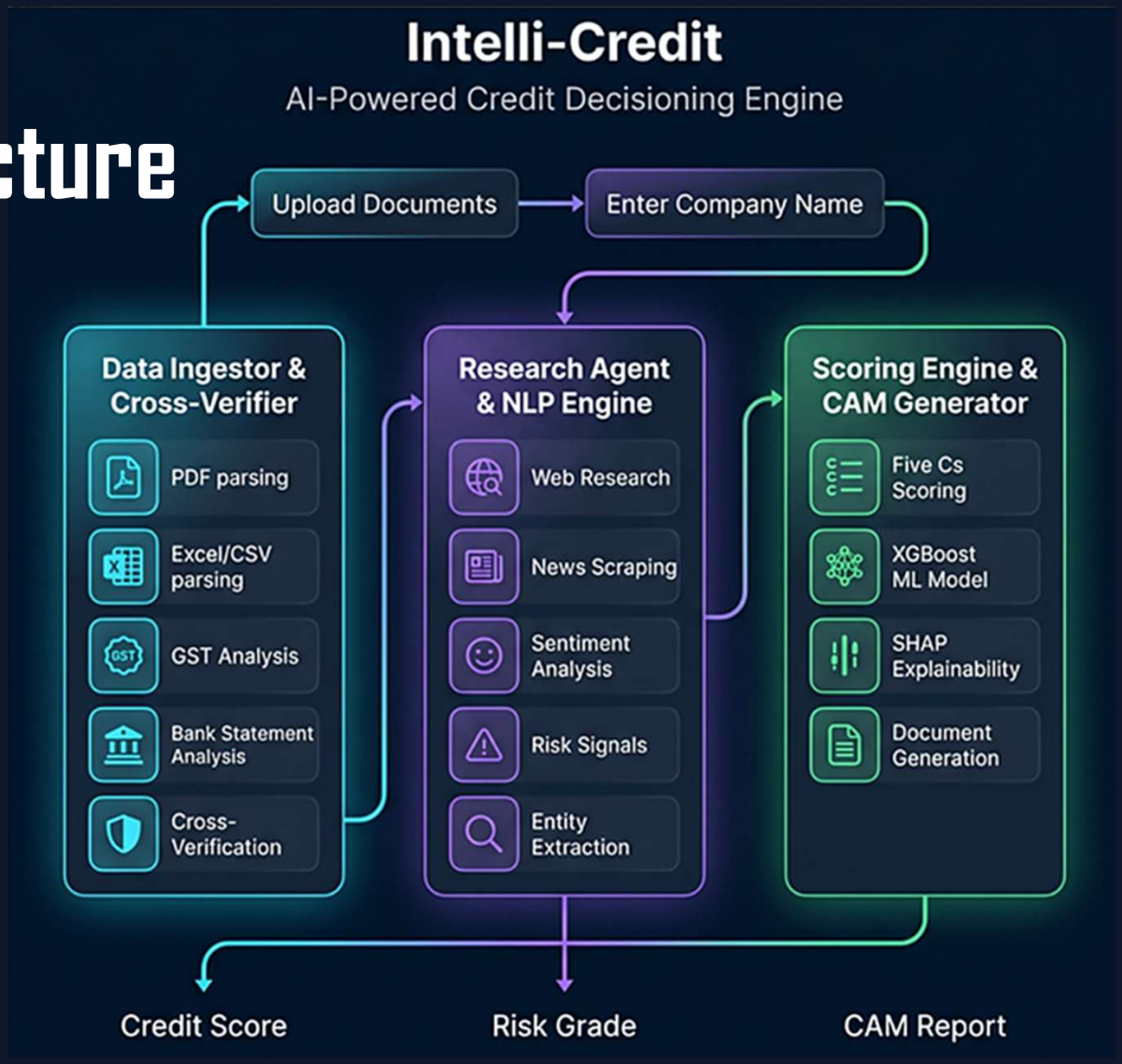
- Credit managers deal with a 'Data Paradox' more data, yet weeks to process a single application.
- Stitching disparate structured data (GST, Bank Statements) and unstructured data (Annual Reports).
- Manual processes miss critical 'early warning signals' buried in unstructured text.
- Primary insights (site visits) and external intelligence (MCA, Legal) remain siloed.
- Result: Slow, manual and bias-prone credit appraisal.

Our Solution: Intelli-Credit

- An AI-Native workflow automating corporate credit decisions end-to-end.
- Ingests multi-source unstructured & structured data seamlessly.
- Performs deep 'web-scale' secondary research on promoters, sector headwinds and litigation.
- Synthesizes primary due diligence (Credit Officer notes) to instantly adjust risk scores.
- Generates a structured, committee-ready Credit Appraisal Memo (CAM) with Explainable AI.

Technical Architecture

- Platform processes uploaded financial documents, verifies financial data, enriches it with external intelligence using NLP.
- Applies machine learning-based credit scoring.
- System outputs an explainable credit score, risk grade, and an automated CAM report to support faster and more reliable lending decisions.



The 7-Step AI Pipeline

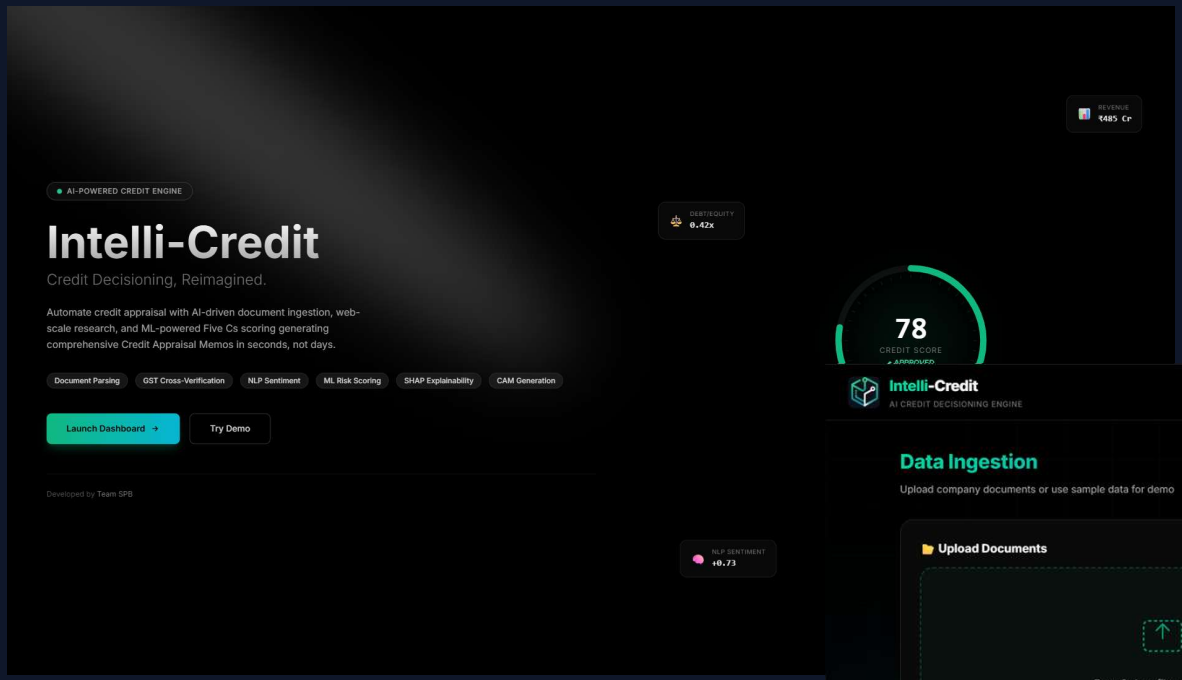
- Financial documents are ingested and cross-verified using GST and bank data.
- External research and NLP identify risk signals, after which machine learning models and the Five Cs framework evaluate creditworthiness.
- The system then generates a credit score, risk grade and an automated CAM report for decision support.

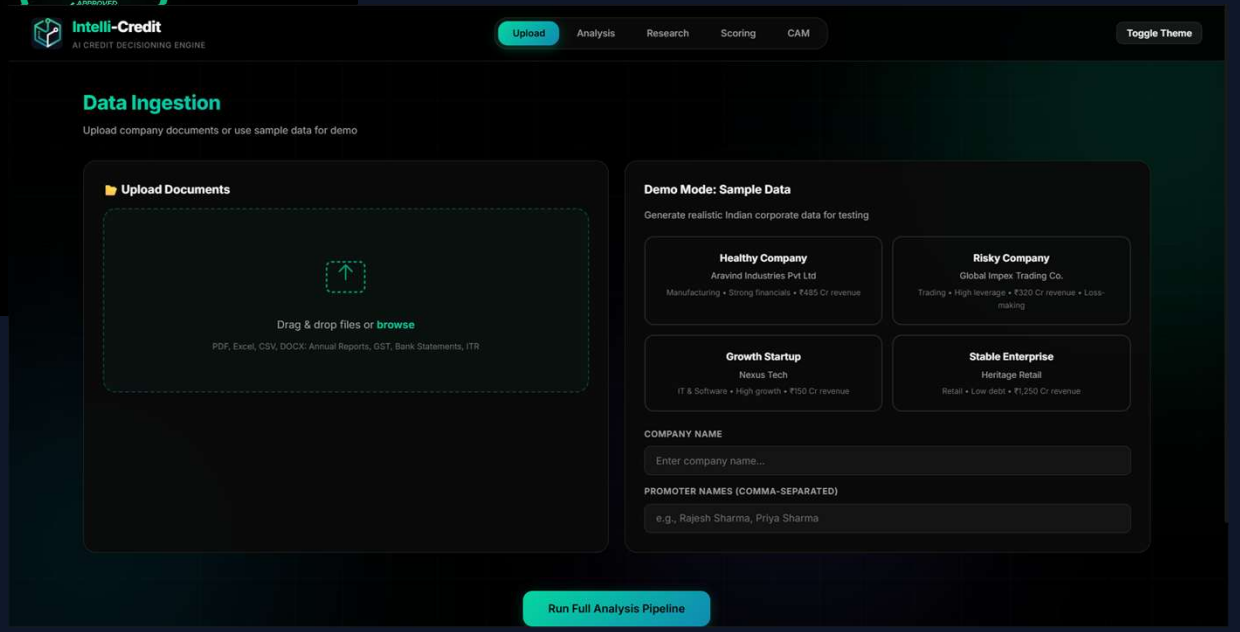


The Three Pillars of Intelli-Credit

- 1. High-Latency Data Ingestor: Parses PDF reports and cross-leverages GST vs Bank Statements to detect 'circular trading'.
- 2. Research Agent ('Digital Credit Manager'): Web-scale crawling for news, MCA filings, and e-Courts litigation.
- 3. Recommendation Engine: Predicts limit/rate via ML and generates the Five Cs CAM.
- Primary Insight Integration: Qualitative factory/site visit notes dynamically adjust the AI model.

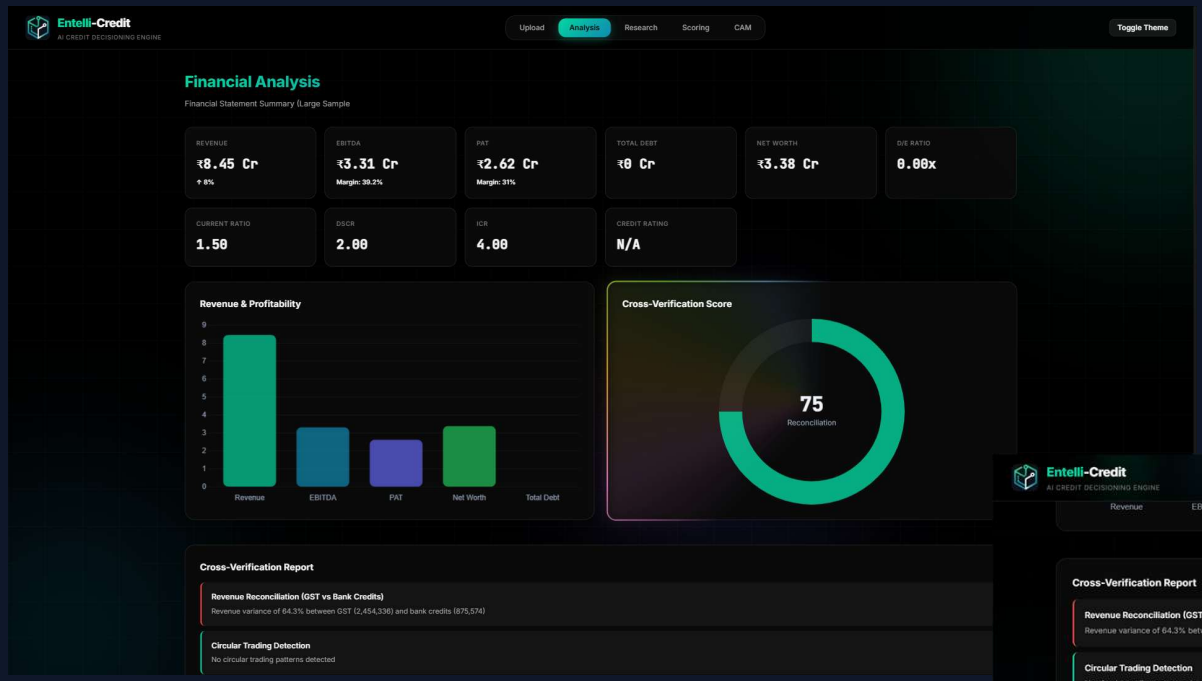
User Interface Landing

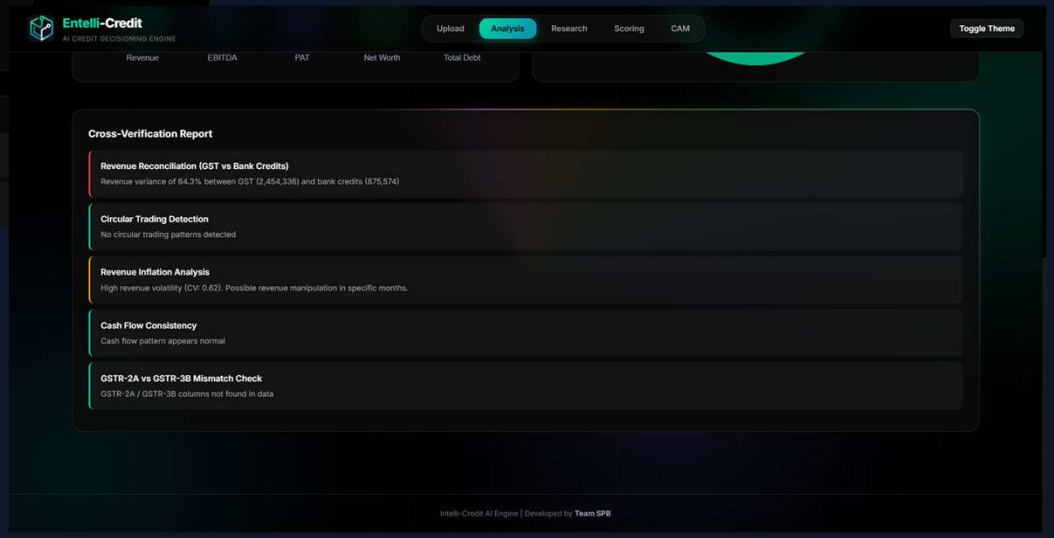




- Clean, modern interface for the Credit Manager.
- Accepts multi-format, unstructured and structured uploads.

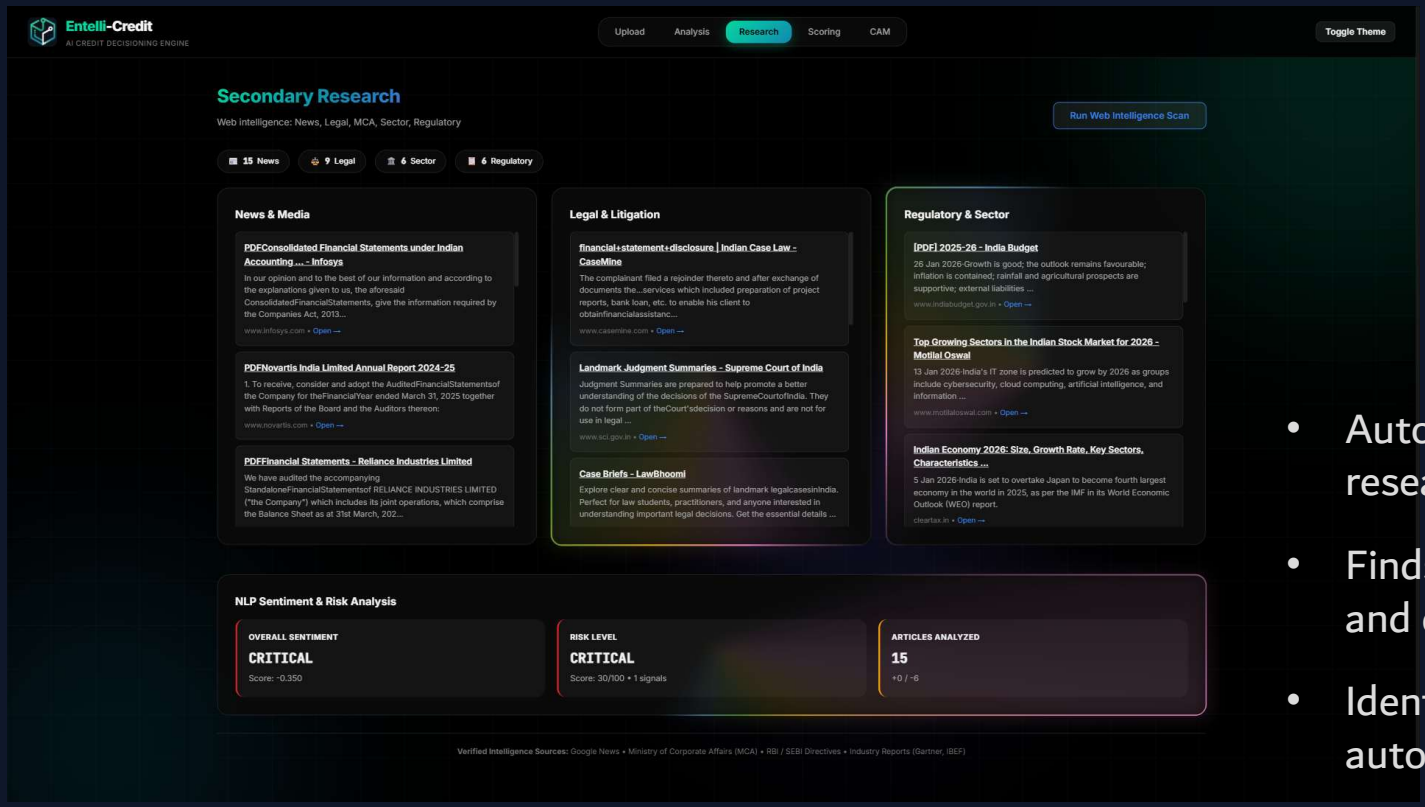
Pillar 1: Data Ingestor & Synthesis





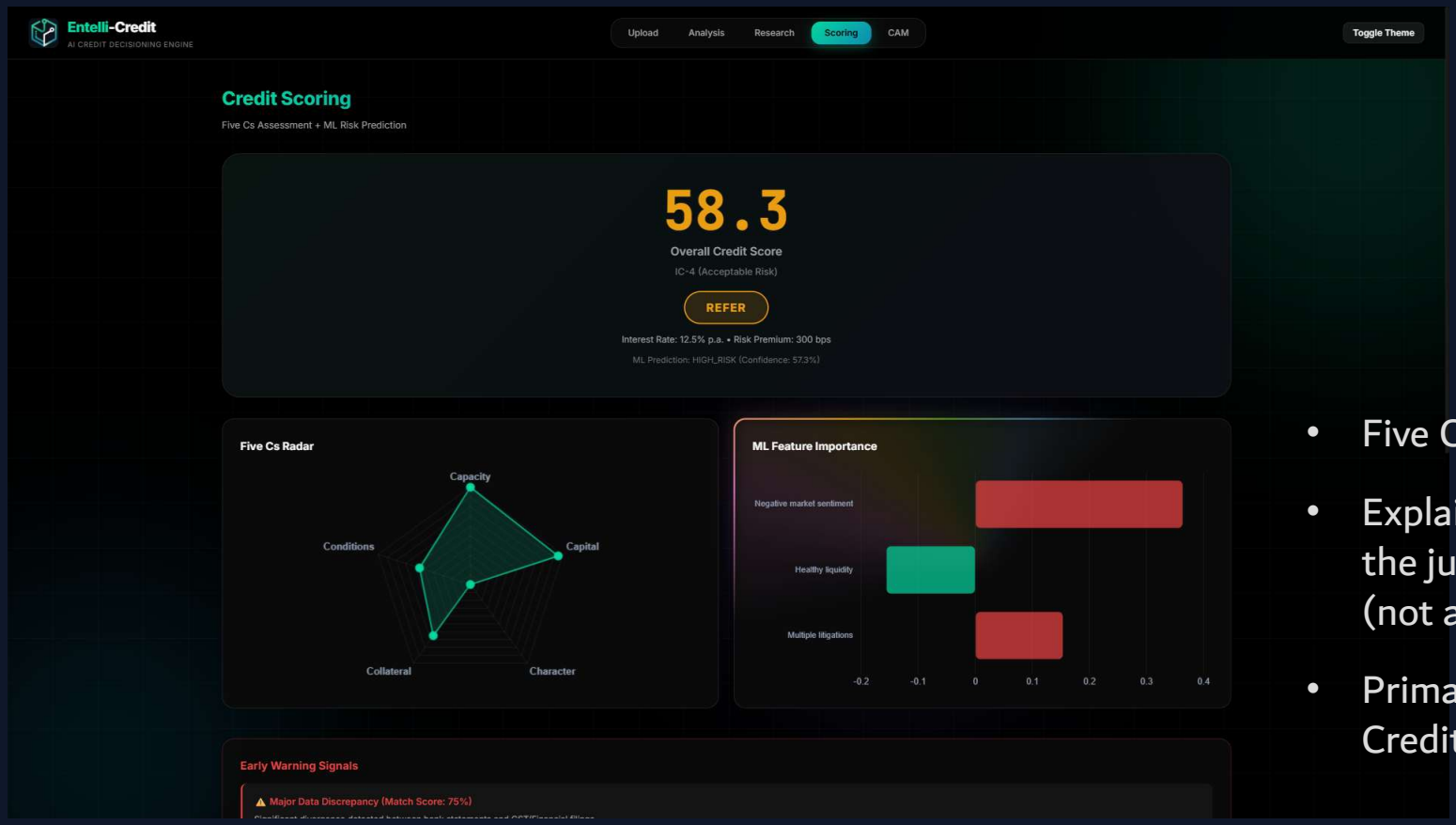
- Cross-leverages GST returns against Bank Statements.
- Specifically designed to identify 'circular trading' or revenue inflation.
- Extracts data from messy Indian context PDFs.

Pillar 2: Research Agent



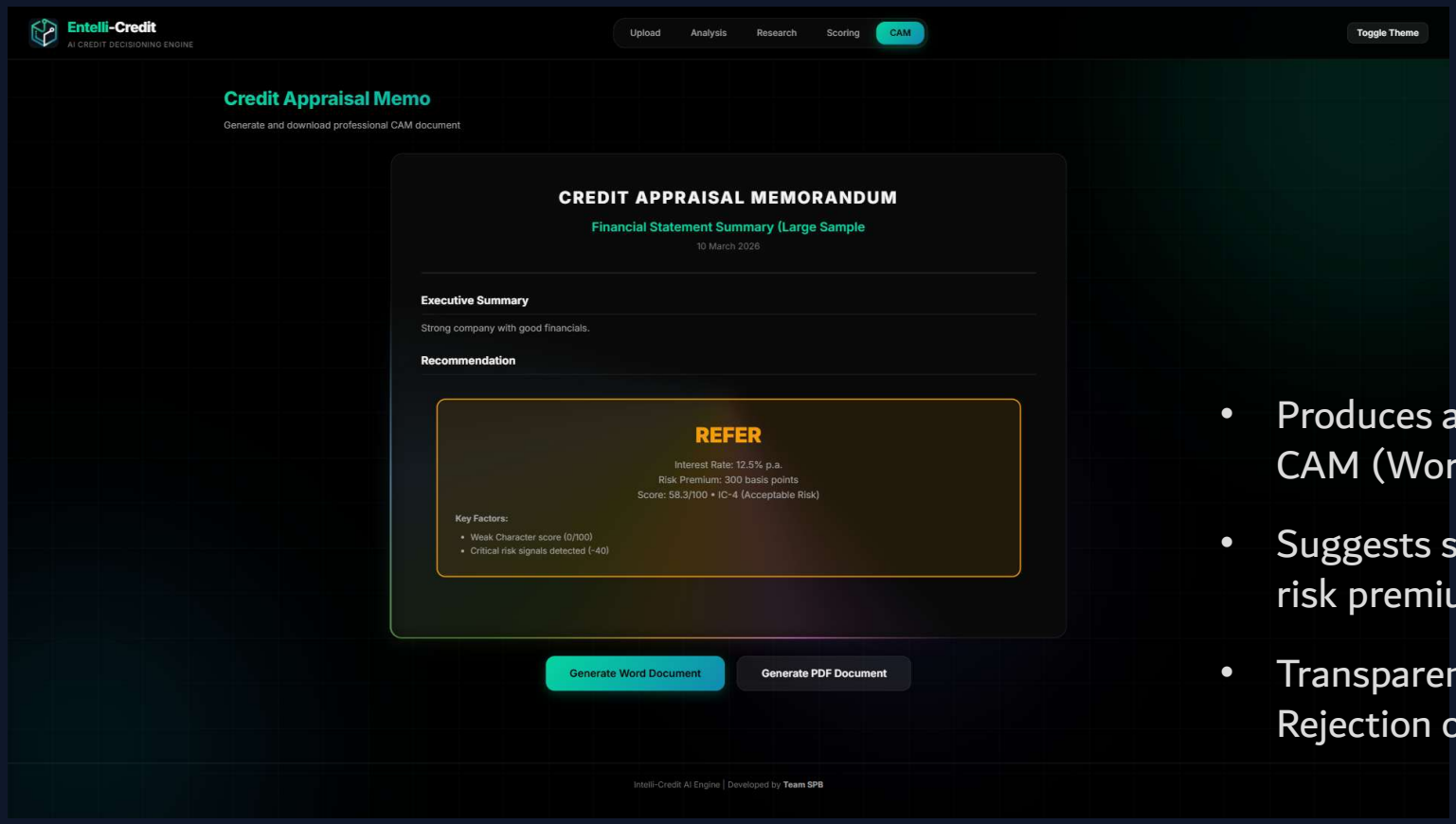
- Automated 'web-scale' secondary research.
- Finds local news, sector headwinds, and e-Court litigation history.
- Identifies 'early warning signals' automatically.

Pillar 3: Recommendation Engine



- Five Cs of Credit breakdown.
- Explainable AI: SHAP charts 'walk the judge through' the ML logic (not a black box).
- Primary Due Diligence: Input for Credit Officer site-visit nuances.

Final Output: Auto-CAM Generator



- Produces a professional, structured CAM (Word/PDF).
- Suggests specific loan amounts and risk premiums.
- Transparent explanations for Rejection or Approval limits.

Technology Stack

- Frontend: HTML5, CSS3, Vanilla JS, Chart.js
- Backend: Python, Flask REST API
- Data Parsing: PyMuPDF, pdfplumber, pandas
- AI/ML Engine: XGBoost, SHAP, scikit-learn
- NLP / LLM: TextBlob, OpenAI GPT-4 API (for synthesis & sentiment)
- Document Generation: python-docx, reportlab

Meeting the Core Evaluation Criteria

- Extraction Accuracy: High-fidelity parsing of Indian-context PDFs and CSVs.
- Research Depth: Automated live-web search capturing local litigation & headwinds.
- Explainability: Complete transparency preventing the 'black box' syndrome via SHAP.
- Indian Context: Handles GST vs Bank reconciliations directly.


Thank You.